Cross-Fitted Clustered Conformal Prediction

October 23, 2026

Abstract

In classification, standard conformal prediction guarantees that prediction sets cover the true label with a chosen probability overall, but the coverage of individual classes is not guaranteed. Classwise conformal prediction guarantees coverage for each class by calibrating a separate threshold for each, but these thresholds can be unstable when few calibration examples are available per class, a common challenge when the calibration set is limited and the number of classes is large. Clustered conformal prediction [3] pools similar classes into clusters to share calibration data, but must split its limited calibration budget between learning the clusters and computing thresholds. We propose cross-fitted clustered conformal prediction (CCCP), which learns the clusters from cross-fitted training scores, leaving the entire calibration set for threshold estimation. The cross-fitted models can also be reused as an ensemble predictor, improving predictive performance and reducing prediction-set size without additional model training. On three image-classification benchmarks (including a controlled unbalanced variant) with 101–397 classes, CCCP achieves the lowest class-conditional coverage gap among the evaluated methods across the calibration regimes considered.

1 Introduction

Machine learning classifiers are increasingly deployed in high-stakes settings like medical diagnosis, autonomous driving, and legal decision-making where reliable uncertainty quantification is essential. In addition to point predictions, it is important to also construct prediction sets with desired coverage levels. However, machine learning classifiers are known to produce overconfident predictions, and their softmax probabilities do not necessarily reflect the true likelihood of being correct. Existing approaches to uncertainty quantification such as Bayesian neural networks and Monte Carlo dropout estimate predictive uncertainty through probabilistic modeling or variation across model predictions, but do not generally provide distribution-free, model-agnostic, finite-sample coverage guarantees.

Conformal prediction [1, 5] offers a fundamentally different approach. Rather than outputting a single class label, it constructs a *prediction set*—a set of candidate labels designed to contain the true label with user-specified probability $1 - \alpha$. Consider classification into C classes with a pretrained classifier f, and let $s(x, y)$ be a nonconformity score for class y given input x, with larger values indicating greater nonconformity. Split conformal prediction calibrates a threshold $\hat{q}$ using the k-th smallest nonconformity score on a held-out calibration set $\mathcal{D}_{\text{cal}}$ of size n_{cal}, where $k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$ and $\hat{q} = +\infty$ if $k > n_{\text{cal}}$. The prediction set is then defined as $\mathcal{C}_\alpha(x) = \{y : s(x, y) \leq \hat{q}\}$. Under exchangeability, this procedure provides the finite-sample marginal coverage guarantee

$$\mathbb{P}\big(Y \in \mathcal{C}_\alpha(X)\big) \geq 1 - \alpha. \tag{1}$$

This guarantee holds without requiring modifications to the underlying classifier [1, 5]. The prediction set is also adaptive: it tends to be smaller when the classifier is confident about a few candidate classes and larger when the classifier is uncertain among many classes.

However, in standard conformal prediction, the marginal guarantee (1) averages over the entire data distribution and can mask substantial disparities across subpopulations: a predictor may achieve the desired overall coverage while systematically under-covering difficult-to-predict cases and over-covering easier ones. Various extensions of conformal prediction have been proposed to characterize or improve coverage conditional on the input features X or the label Y. Mondrian conformal prediction [13] provides a framework for group-conditional coverage by partitioning the feature or label space into disjoint groups and calibrating separately within each group. In regression problems with continuous Y, a common objective is to achieve covariate-conditional coverage, which requires the prediction set to achieve the desired coverage level at each input value x:

$$\mathbb{P}\big(Y \in \mathcal{C}_\alpha(X) \mid X = x\big) \geq 1 - \alpha \qquad \text{for all } x. \tag{2}$$

A substantial body of research has been motivated by this objective [17, 18]. In classification problems, the objective is often to improve the class-conditional coverage so that the desired coverage is achieved separately for each class:

$$\mathbb{P}\big(Y \in \mathcal{C}_\alpha(X) \mid Y = y\big) \geq 1 - \alpha \qquad \text{for every class } y, \tag{3}$$

which is the focus of this paper.

Classwise conformal prediction [2]—a special case of Mondrian CP that uses each class as its own group—achieves the class-conditional coverage in (3) by calibrating a separate threshold $\hat{q}_y$ for each class y using only the calibration examples from that class. However, classwise conformal prediction requires a sufficient number of calibration examples in each class to estimate its threshold reliably. In many real-world settings, where the sample is limited and the number of classes is large, some classes may have too few calibration examples for reliable threshold estimation. Moreover, when the number of calibration examples in class y, $n_{\text{cal},y}$, falls below the floor

$$n_{\text{cal},y} \;\geq\; \lceil 1/\alpha \rceil - 1, \tag{4}$$

the required order statistic does not exist, and the corresponding threshold is set to $+\infty$, causing every prediction set to include class y. More recent studies explore alternative approaches to improve classwise coverage. RC3P [14] selectively applies class-wise thresholds only where per-class data suffices. Ding et al. [15] interpolate between standard and classwise thresholds for long-tailed settings. Bhattacharyya et al. [16] propose macro-coverage—the unweighted average of class-conditional coverages—as an alternative guarantee between marginal and class-conditional.

Clustered conformal prediction [3] takes a different approach by grouping the C classes into M clusters based on their score distributions and calibrating a single threshold per cluster. By pooling calibration examples across similar classes, it can increase the effective sample size available for threshold estimation compared with classwise conformal prediction. The full procedure is described in Section 2.1. However, the clustering procedure uses a significant proportion of the calibration set to learn the clusters. These samples can no longer be used for calibrating the threshold, which can be particularly problematic when calibration data are scarce. To overcome this limitation, we propose cross-fitted clustered conformal prediction (CCCP), which learns cluster assignments from cross-fitted scores on the training data rather than from the calibration set. Inspired by cross-fitting techniques in conformal inference [6] and causal inference [8, 7], CCCP partitions the training set into K folds, trains a subfold model on $K-1$ folds, and computes scores on the held-out fold. The

resulting cross-fitted scores are used to learn the class clusters, allowing the entire calibration set to be reserved for threshold estimation. Under the appropriate exchangeability assumptions, this separation preserves the cluster-conditional coverage guarantee while avoiding the calibration-data split required by the original clustered CP procedure. Moreover, the K subfold models trained during cross-fitting can be repurposed to construct an ensemble predictor (EnsCCCP) at no additional training cost. By aggregating predictions across multiple models, EnsCCCP can improve predictive accuracy and empirically reduce prediction-set sizes, while retaining the same coverage guarantee as CCCP.

Our main contributions are summarized as follows:

1. We propose cross-fitted clustered conformal prediction (CCCP), which learns class clusters from cross-fitted training scores rather than the calibration set, allowing the full calibration set to be used for threshold estimation while preserving cluster-conditional coverage under appropriate assumptions.

2. We develop EnsCCCP, an ensemble extension that reuses the cross-fitted subfold models without additional training. Experiments demonstrate its potential to improve predictive accuracy and reduce prediction-set sizes.

3. We conduct a comprehensive empirical evaluation of class-conditional coverage and prediction-set efficiency across four large-scale classification datasets, comparing CCCP and EnsCCCP with existing conformal prediction methods under different calibration regimes.

The remainder of this paper is organized as follows. Section 2 introduces the clustered conformal prediction framework and presents the proposed CCCP and EnsCCCP methods. Section 3 establishes their theoretical coverage guarantees. Section 4 evaluates the proposed methods on four large-scale classification datasets under various calibration regimes. Finally, Section 5 concludes the paper.

2 Method

2.1 Preliminaries: Clustered Conformal Prediction

Clustered conformal prediction (clustered CP), introduced by Ding et al. [3], pools classes with similar score distributions. Rather than calibrating one threshold per class, it groups the C classes into M clusters and calibrates one threshold per cluster. Each cluster aggregates calibration points from many classes, so its effective sample size comfortably clears the floor (4) even when individual per-class counts do not. The coverage guarantee weakens from class-conditional to cluster-conditional—a middle ground between marginal and class-conditional coverage—but this trade-off is favorable in the small-sample regime where classwise CP fails.

The full procedure is given in Algorithm 1. The calibration set $\mathcal{D}_{\text{cal}}$ is randomly split into a clustering set $\mathcal{D}_1$ (fraction γ) and a proper-calibration set $\mathcal{D}_2$ (fraction $1-\gamma$), with $\mathcal{D}_1$ and $\mathcal{D}_2$ disjoint so that the clustering is fixed before calibration. Each non-rare class is then summarized by a fixed-length quantile embedding: the vector of empirical quantiles of that class's nonconformity scores at the nine decile levels $0.1, 0.2, \ldots, 0.9$, computed on $\mathcal{D}_1$. This nine-number summary captures the location and shape of a class's score distribution in a representation whose dimension does not depend on n_y, so that two classes with similarly distributed scores map to nearby points in $\mathbb{R}^9$. Weighted k-means is then run on these embeddings, with class y weighted by $\sqrt{n_y^1}$ (the square root of its clustering-set count) so that better-estimated classes exert proportionally more influence,

Algorithm 1 Clustered Conformal Prediction [3]

Require: confidence level α, clustering fraction γ

Input: training set $\mathcal{D}_{\text{train}}$, calibration set $\mathcal{D}_{\text{cal}}$, new data point x_{new}

1: train classifier f on $\mathcal{D}_{\text{train}}$; compute nonconformity scores $s(x, y)$
2: randomly split $\mathcal{D}_{\text{cal}}$ into a clustering set $\mathcal{D}_1$ (fraction γ) and a proper-calibration set $\mathcal{D}_2$
3: mark classes with fewer than n_α examples in $\mathcal{D}_1$ as rare
4: compute the number of clusters M
5: compute the quantile embedding z_y for each non-rare class y from its scores in $\mathcal{D}_1$
6: run weighted k-means on $\{z_y\}$ to obtain cluster assignment $\hat{h}$; assign all rare classes to a null cluster
7: for each cluster m, set threshold $\hat{q}_m$ from the pooled scores of its member classes in $\mathcal{D}_2$; set $\hat{q}_{\text{null}}$ to the marginal threshold over all of $\mathcal{D}_2$
8: **return** prediction set $\mathcal{C}_\alpha(x_{\text{new}}) = \{\, y : s(x_{\text{new}}, y) \le \hat{q}_{\hat{h}(y)} \,\}$

producing an assignment $\hat{h}$ of classes to M clusters. Classes with fewer than $n_\alpha = 1/\min\{\alpha, 0.1\} - 1$ points in $\mathcal{D}_1$ ($n_\alpha = 9$ at $\alpha = 0.1$) are too rare to embed reliably; they are placed in a separate null cluster and given the marginal (all-class) threshold rather than a cluster threshold. Finally, treating each cluster as a super-class, one pools on $\mathcal{D}_2$ the true-label scores of all its member classes and sets $\hat{q}_m$ to the $\lceil (1-\alpha) \rceil$ quantile of the N_m pooled scores. Because $\hat{h}$ is a function of $\mathcal{D}_1$ alone and $\mathcal{D}_1 \perp \mathcal{D}_2$, each cluster is a valid Mondrian group, giving cluster-conditional coverage $\mathbb{P}(Y \in \mathcal{C}_\alpha(X) \mid \text{cluster}(Y) = m) \ge 1 - \alpha$ for every non-null cluster.

The number of clusters M must be large enough to group classes with genuinely similar score distributions, yet small enough that each cluster pools sufficient calibration points to clear the floor (4). We use an initial estimate with post-hoc merging: set an initial $M = \max(2, \lfloor N_{\text{cal,nn}}/300 \rfloor)$, where $N_{\text{cal,nn}}$ is the total calibration count across non-rare classes, then iteratively merge the smallest cluster (by calibration count) into its nearest neighbor in embedding space until every surviving cluster has at least 300 calibration points. This enforces a hard minimum budget per cluster at the cost of a data-dependent final M.

Clustering and calibration are carved from the same held-out sample: a fraction γ goes to clustering and $1 - \gamma$ to calibration, with the heuristic $\gamma = C_{\text{NR}}/(C_{\text{NR}} + 75)$, where C_{NR} is the number of non-rare classes (for $C = 100$ fully populated classes, $\gamma \approx 4/7$). Every point spent on clustering is therefore unavailable for calibration: the proper-calibration set shrinks to $(1-\gamma)\, n$, i.e. to $\approx (1-\gamma)\, n_{\text{cal}}/C$ per class. When n_{cal}/C is small—the very regime clustered CP targets—this can push the per-cluster counts back toward the floor (4), undermining the method exactly where it is needed. CCCP removes this trade-off.

2.2 Cross-Fitted Clustered Conformal Prediction (CCCP)

In clustered CP the clustering set and the calibration set compete for one scarce budget. With $\gamma \approx 0.57$ the proper-calibration set is only $\approx 0.43\, n_{\text{cal}}/C$ per class, which can dip below the floor (4) exactly when n_{cal}/C is small. CCCP takes a different approach: it extracts clustering information from the training set itself via cross-fitting, leaving the calibration set entirely for calibration—no γ-split and no separate clustering model.

Start from the same partition as clustered CP: a training set $\mathcal{D}_{\text{train}}$ and a calibration set $\mathcal{D}_{\text{cal}}$. Train the prediction model f on $\mathcal{D}_{\text{train}}$ exactly as in the single-model baseline. To obtain out-of-sample scores for clustering, partition $\mathcal{D}_{\text{train}}$ into K class-stratified folds $\mathcal{F}_1, \ldots, \mathcal{F}_K$ and, for each fold $\mathcal{F}_k$, train a subfold model g_k on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$ and score $\mathcal{F}_k$ with g_k. The pooled cross-fitted

Algorithm 2 Cross-Fitted Clustered Conformal Prediction (CCCP)

Require: confidence level α, number of cross-fitting folds K

Input: training set $\mathcal{D}_{\text{train}}$, calibration set $\mathcal{D}_{\text{cal}}$, new data point x_{new}

train prediction model f on $\mathcal{D}_{\text{train}}$; compute nonconformity scores $s(x, y)$
partition $\mathcal{D}_{\text{train}}$ into K class-stratified folds $\mathcal{F}_1, \ldots, \mathcal{F}_K$
for $k = 1, \ldots, K$ **do**
    train subfold model g_k on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$; compute $s_k(x, y)$ for $(x, y) \in \mathcal{F}_k$
pool cross-fitted clustering scores: $\{(x,\ y,\ s_k(x, y)) : (x, y) \in \mathcal{F}_k,\ k = 1, \ldots, K\}$
mark classes with fewer than n_α examples in $\mathcal{D}_{\text{train}}$ as rare
compute the number of clusters M
compute the quantile embedding z_y for each non-rare class y from its cross-fitted scores
run weighted k-means on $\{z_y\}$ to obtain cluster assignment $\hat{h}$; assign all rare classes to a null cluster
for each cluster m, set threshold $\hat{q}_m$ from the pooled scores of its member classes in $\mathcal{D}_{\text{cal}}$ (scored by f); set $\hat{q}_{\text{null}}$ to the marginal threshold over all of $\mathcal{D}_{\text{cal}}$
return prediction set $\mathcal{C}_\alpha(x_{\text{new}}) = \{\, y : s(x_{\text{new}}, y) \le \hat{q}_{\hat{h}(y)} \,\}$

scores cover all of $\mathcal{D}_{\text{train}}$ with every point scored out of sample, and serve as the clustering set. The calibration set $\mathcal{D}_{\text{cal}}$ is scored by the prediction model f—out of sample, since f was trained on $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{cal}}$ is disjoint—and used in its entirety for threshold computation.

Procedure. The full procedure is given in Algorithm 2.

1. **Cross-fit**: partition $\mathcal{D}_{\text{train}}$ into K folds; for each fold $\mathcal{F}_k$, train g_k on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$ and score $\mathcal{F}_k$ out of sample.
2. **Cluster** using the pooled cross-fitted scores on $\mathcal{D}_{\text{train}}$: build quantile embeddings and k-means groups as in clustered CP—with no γ-split.
3. **Calibrate** using $\mathcal{D}_{\text{cal}}$ scored by f: for each cluster m set $\hat{q}_m$ from the pooled $\mathcal{D}_{\text{cal}}$ scores—with no γ-split.
4. **Predict**: score x_{new} with f and include y iff $s(x_{\text{new}}, y) \le \hat{q}_{\hat{h}(y)}$.

Effect on the effective calibration set. Writing n_{cal} for the total size of $\mathcal{D}_{\text{cal}}$ and n_{tr} for the total size of $\mathcal{D}_{\text{train}}$, the data available to each role is:

	training size	clustering size	calibration size
Clustered CP (γ-split)	n_{tr}	$\gamma\, n_{\text{cal}}$	$(1-\gamma)\, n_{\text{cal}}$
CCCP	n_{tr}	n_{tr}	n_{cal}

CCCP gives the calibration set the full n_{cal} without a γ-split. The clustering set is $\mathcal{D}_{\text{train}}$, which is typically much larger than the γ-fraction of $\mathcal{D}_{\text{cal}}$ used by clustered CP. The price is computational: CCCP trains K additional subfold models.

The prediction model f is trained on $\mathcal{D}_{\text{train}}$ alone and is therefore independent of $\mathcal{D}_{\text{cal}}$. Because each subfold model g_k is trained on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$—also independent of $\mathcal{D}_{\text{cal}}$—the clustering map $\hat{h}$ is a function of $\mathcal{D}_{\text{train}}$ alone. Both $\hat{h}$ and f are therefore independent of $\mathcal{D}_{\text{cal}}$, and within each cluster the $\mathcal{D}_{\text{cal}}$ scores and the test score are exchangeable under f, giving clean cluster-conditional coverage $\ge 1 - \alpha$ without any dependence subtlety.

Algorithm 3 CCCP Ensemble (EnsCCCP)

Require: confidence level α, number of cross-fitting folds K

Input: training set $\mathcal{D}_{\text{train}}$, calibration set $\mathcal{D}_{\text{cal}}$, new data point x_{new}

partition $\mathcal{D}_{\text{train}}$ into K class-stratified folds $\mathcal{F}_1, \ldots, \mathcal{F}_K$
for $k = 1, \ldots, K$ **do**
    train subfold model g_k on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$; compute $s_k(x, y)$ for $(x, y) \in \mathcal{F}_k$
pool cross-fitted clustering scores: $\mathcal{D}_{\text{clus}} = \bigcup_{k=1}^{K} \{(x,\, y,\, s_k(x, y)) : (x, y) \in \mathcal{F}_k\}$
mark classes with fewer than n_α examples in $\mathcal{D}_{\text{clus}}$ as rare
compute the number of clusters M
compute the quantile embedding z_y for each non-rare class y from its scores in $\mathcal{D}_{\text{clus}}$
run weighted k-means on $\{z_y\}$ to obtain cluster assignment $\hat{h}$; assign all rare classes to a null cluster
define ensemble predictor $f_{\text{ens}}(x) = \frac{1}{K} \sum_{k=1}^{K} g_k(x)$; set $s_{\text{ens}}(x, y) = 1 - (f_{\text{ens}})_y(x)$
for each cluster m, set threshold $\hat{q}_m$ from the pooled ensemble scores of its member classes in $\mathcal{D}_{\text{cal}}$ (scored by f_{ens}); set $\hat{q}_{\text{null}}$ to the marginal threshold over all of $\mathcal{D}_{\text{cal}}$
return prediction set $\mathcal{C}_\alpha(x_{\text{new}}) = \{\, y : s_{\text{ens}}(x_{\text{new}}, y) \le \hat{q}_{\hat{h}(y)} \,\}$

2.3 CCCP Ensemble (EnsCCCP)

CCCP trains K subfold models solely to produce out-of-sample clustering scores on $\mathcal{D}_{\text{train}}$; at prediction time these models are discarded and the single prediction model f is used. EnsCCCP repurposes the same subfold models as an ensemble predictor: averaging their softmax outputs produces a stronger classifier than any individual model, which in turn yields smaller prediction sets while preserving the same clustering and validity guarantee.

EnsCCCP uses the same partition and cross-fitting structure as CCCP (Section 2.2). The K subfold models $g_1, \ldots, g_K$ are trained on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$ exactly as before, and their cross-fitted scores on $\mathcal{D}_{\text{train}}$ are used to cluster the classes identically. The difference is in calibration and prediction: instead of the single model f, EnsCCCP defines the ensemble predictor

$$f_{\text{ens}}(x) \;=\; \frac{1}{K} \sum_{k=1}^{K} g_k(x), \tag{5}$$

where $g_k(x)$ denotes the softmax output vector of the k-th subfold model. Both $\mathcal{D}_{\text{cal}}$ and the test point are scored by f_{ens}, with nonconformity score $s_{\text{ens}}(x, y) = 1 - (f_{\text{ens}})_y(x)$. Because no subfold model was trained on $\mathcal{D}_{\text{cal}}$, these calibration scores are out of sample. The prediction model f is not used and need not be trained if only the ensemble variant is desired.

Procedure. The full procedure is given in Algorithm 3.

1. **Cross-fit and cluster**: identical to CCCP (Steps 1–8 of Algorithm 2).
2. **Calibrate**: score $\mathcal{D}_{\text{cal}}$ with the ensemble f_{ens}; for each cluster m set $\hat{q}_m$ from the pooled ensemble scores—with no γ-split.
3. **Predict**: score x_{new} with f_{ens} and include y iff $s_{\text{ens}}(x_{\text{new}}, y) \le \hat{q}_{\hat{h}(y)}$.

EnsCCCP reuses the K subfold models that CCCP already trains, so it incurs no additional training cost. The ensemble is typically a more accurate predictor than the single model f, and this higher accuracy translates directly into smaller prediction sets.

The costs are twofold. First, prediction requires K forward passes instead of one. Second, the clustering is formed from individual subfold models' cross-fitted scores on $\mathcal{D}_{\text{train}}$, not from the ensemble's scores, so the cluster assignment may be suboptimal for the ensemble's score scale.

Each subfold model g_k is trained on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$, independent of $\mathcal{D}_{\text{cal}}$, so the ensemble f_{ens} is also independent of $\mathcal{D}_{\text{cal}}$. The clustering map $\hat{h}$ is a function of $\mathcal{D}_{\text{train}}$ alone (as in CCCP). Both $\hat{h}$ and f_{ens} are therefore independent of $\mathcal{D}_{\text{cal}}$, and within each cluster the ensemble-scored $\mathcal{D}_{\text{cal}}$ points and the test point are exchangeable, giving cluster-conditional coverage $\geq 1 - \alpha$.

2.4 Comparison of methods

Table 1 summarizes the conformal prediction methods considered in this paper. Standard and Classwise require a single model; Clustered adds no models but sacrifices calibration data to a γ-split. CCCP eliminates the split by cross-fitting within the training set, at the cost of K additional subfold models; EnsCCCP repurposes those same subfold models as an ensemble predictor, trading K forward passes at prediction time for smaller prediction sets. EnsStd applies standard conformal prediction on top of the same ensemble predictor, providing a marginal baseline that isolates the effect of clustering from the effect of ensembling.

Table 1: Comparison of conformal prediction methods. All methods use the same training and calibration sets for fair comparison. Training cost is the number of models trained; prediction cost is the number of forward passes per new data point.

Method	Training cost	Prediction cost	Clustering size	Proper-cal size	Coverage Guarantee
Standard	1 model	1 pass	—	n_{cal}	marginal
Classwise	1 model	1 pass	—	n_{cal}	class-conditional
Clustered	1 model	1 pass	γn_{cal}	$(1-\gamma)n_{\text{cal}}$	cluster-conditional
CCCP	$1+K$ models	1 pass	n_{tr}	n_{cal}	cluster-conditional
EnsCCCP	K models	K passes	n_{tr}	n_{cal}	cluster-conditional

3 Theoretical Analysis

3.1 Cluster-conditional coverage

Both Clustered CP and CCCP satisfy the cluster-conditional guarantee of Ding et al. [3], which holds whenever the clustering map is independent of the calibration data.

Proposition 1 (Cluster-conditional coverage [3])**.** *Let $\hat{h} : \{1, \ldots, C\} \to \{1, \ldots, M, \text{null}\}$ be a clustering map independent of the proper-calibration set $\mathcal{D}_2$ that assigns each class to a cluster, and let $\hat{q}_m$ be the $\lceil (1-\alpha) \rceil$ quantile of the calibration scores in cluster m. Then for every non-null cluster m,*

$$\mathbb{P}\big(Y \in \mathcal{C}_\alpha(X) \mid \hat{h}(Y) = m\big) \;\geq\; 1 - \alpha, \tag{6}$$

where $\mathcal{C}_\alpha(X) = \{y : s(X, y) \leq \hat{q}_{\hat{h}(y)}\}$ is the prediction set.

In clustered CP, independence holds because $\hat{h}$ is constructed from the clustering split $\mathcal{D}_1$, which is disjoint from $\mathcal{D}_2$. In CCCP, $\hat{h}$ is constructed entirely from $\mathcal{D}_{\text{train}}$ via cross-fitting, so it is likewise independent of $\mathcal{D}_{\text{cal}}$, and Proposition 1 applies directly with $\mathcal{D}_2 = \mathcal{D}_{\text{cal}}$.

3.2 Approximate class-conditional coverage

Beyond cluster-conditional validity, Ding et al. [3] give an approximate class-conditional guarantee that bounds how far per-class coverage can fall below the target level.

The total-variation distance between two distributions is

$$\mathrm{TV}(P,Q) \;=\; \sup_{A} |P(A) - Q(A)| \;=\; \tfrac{1}{2}\int |p - q|,$$

lying in $[0,1]$: 0 when the two distributions coincide, 1 when their supports are disjoint.

Proposition 2 (Approximate class-conditional coverage [3])**.** *Under the conditions of Proposition 1, suppose that for every pair of classes y, y' assigned to the same non-null cluster,*

$$\mathrm{TV}\big(s(X,y) \mid Y{=}y,\;\; s(X,y') \mid Y{=}y'\big) \;\le\; \epsilon \qquad \textit{whenever} \quad \hat{h}(y) = \hat{h}(y') \neq \text{null}. \tag{7}$$

Then for every class y in a non-null cluster,

$$\mathbb{P}\big(Y \in \mathcal{C}_\alpha(X) \mid Y = y\big) \;\ge\; 1 - \alpha - \epsilon. \tag{8}$$

The class-conditional guarantee therefore degrades linearly in ϵ, the worst-case within-cluster spread of score distributions.

Remark 1 (CCCP and score-scale alignment)**.** *In clustered CP the clusters are formed on the same model's score scale as calibration, so k-means directly minimizes a proxy for the total-variation quantity in* (7)*. In CCCP the clusters are formed on the cross-fitted subfold models' score scales, but ϵ is evaluated on the prediction model f's scale. Because each subfold model g_k is trained on $\mathcal{D}_{\text{train}} \backslash \mathcal{F}_k$ (all but $1/K$ of the training data), its learned representations closely approximate those of f (trained on all of $\mathcal{D}_{\text{train}}$). The score-scale mismatch is therefore small and shrinks as K increases. Empirically, CCCP achieves lower ϵ and better classwise coverage than clustered CP, because the much larger clustering set (n_{tr} vs. $\gamma n_{\text{cal}}/C$) produces more accurate quantile embeddings that more than compensate for any residual scale mismatch.*

4 Experiments

4.1 Datasets

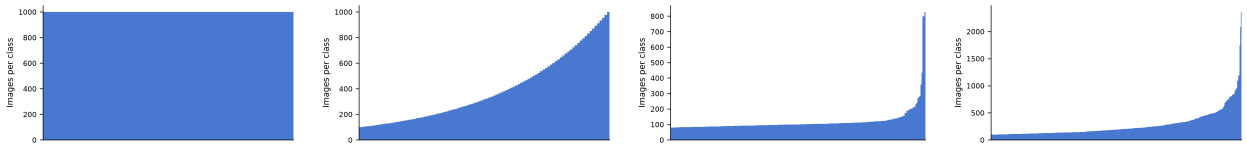
We evaluate on three image-classification datasets: Caltech-256 [10] ($C = 257$, imbalanced, 30,607 images), SUN-397 [11] ($C = 397$, imbalanced, 108,754 images), and Food-101 [12] ($C = 101$, balanced, 101,000 images). To test robustness to class imbalance within a controlled setting, we also create an unbalanced variant of Food-101 by subsampling with an exponential schedule: classes are randomly ordered and assigned counts decaying exponentially from 1,000 to 100 (10:1 imbalance ratio), yielding 39,587 images with the same 101 classes. Figure 1 shows the per-class frequency distribution for each dataset.

4.2 Setup

The base classifier is a ResNet-18 pretrained on ImageNet, fully fine-tuned with Adam ($\mathrm{lr} = 10^{-4}$, batch size 128) and early stopping (patience 3, up to 1000 epochs, best-validation-accuracy checkpoint). We evaluate three nonconformity scores.

Table 2: Summary of datasets used in the experiments. Class balance is the mean count in the rarest 5% of classes divided by the average count per class. This metric is bounded between 0 and 1, where 1 indicates a perfectly balanced distribution and lower values indicate greater class imbalance.

Data set	Classes	Images	$n_{\min}$	$n_{\max}$	Class balance
Food-101	101	101,000	1,000	1,000	1.00
Food-101 (unbal.)	101	39,587	100	1,000	0.27
Caltech-256	257	30,607	80	827	0.68
SUN-397	397	108,754	100	2,361	0.37



(a) Food-101 (b) Food-101 (unbal.) (c) Caltech-256 (d) SUN-397

Figure 1: Number of images per class for each dataset, sorted in increasing order. Food-101 is balanced; Food-101 (unbal.) is an exponentially subsampled variant (10× ratio); the other two are naturally imbalanced.

Homogeneous Prediction Sets (HPS). The HPS score is $s(x, y) = 1 - f_y(x)$, where $f_y(x)$ is the softmax output for class y. A class is included in the prediction set whenever the classifier assigns it sufficiently high probability: $\mathcal{C}_\alpha(x) = \{y : f_y(x) \geq 1 - \hat{q}\}$. The resulting sets are homogeneous in the sense that inclusion depends only on whether a class's softmax probability exceeds the threshold, regardless of how the remaining probability mass is distributed.

Adaptive Prediction Sets (APS). The APS score [4] is constructed by sorting classes in decreasing order of softmax probability and accumulating: $s(x, y) = \sum_{j=1}^{k} f_{\pi_j}(x)$, where π is the permutation that sorts $f(x)$ and $y = \pi_k$. The prediction set includes all classes whose cumulative softmax mass is needed to reach the threshold. APS adapts to the shape of the predictive distribution: when the model concentrates most mass on a few classes, the set is small; when mass is spread diffusely, the set grows.

Regularized Adaptive Prediction Sets (RAPS). RAPS [9] adds a penalty term to the APS score that encourages smaller prediction sets: $s^{\text{RAPS}}(x, y) = s^{\text{APS}}(x, y) + \lambda \max(o(x, y) + 1 - k_{\text{reg}}, 0)$, where $o(x, y)$ is the rank of class y (zero-indexed) in the sorted softmax and $\lambda, k_{\text{reg}} \geq 0$ are hyperparameters. Classes ranked beyond k_{reg} receive a growing additive penalty, making them less likely to enter the prediction set. We use $\lambda = 0.01$ and $k_{\text{reg}} = 5$.

The target level is $\alpha = 0.1$ and we use $K = 10$ cross-fitting folds. Each experiment consists of $T = 20$ independent iterations with different seeds. In each iteration, n_{train} images are drawn via class-stratified sampling as the training set; the remainder forms the rest pool. We use $n_{\text{train}} =$ 10,000 for all datasets. A stratified calibration set of size n_{cal} is drawn from the rest pool and the remaining images serve as the test set. We sweep n_{cal}/C from 4 to 40 in steps of 2 and report results averaged over the $T = 20$ iterations.

4.3 Results

We first compare four conformal methods—Standard, Classwise, Clustered, and CCCP—to evaluate class-conditional coverage under limited calibration data. Table 3 reports detailed results at $n_{\text{cal}}/C \in \{10, 30\}$ and Figure 2 shows CovGap across the full sweep.

Table 3: Results on four datasets with HPS, APS, and RAPS scores ($\alpha = 0.1$, mean±se over 20 iterations, $n_{\text{train}} = 10{,}000$). Marg. = marginal coverage (target 0.90); CovGap (%) = $\frac{1}{C}\sum_y |\widehat{\text{cov}}_y - (1-\alpha)| \times 100$, the mean absolute per-class coverage deviation (lower is better); AvgSize = mean prediction-set cardinality (lower is better); C_{NR} = number of non-rare classes. Bold marks the smallest CovGap in each group.

			$n_{\text{cal}}/C = 10$				$n_{\text{cal}}/C = 30$			
Data set	Score	Method	Marg.	CovGap	AvgSize	C_{NR}	Marg.	CovGap	AvgSize	C_{NR}
Food-101	HPS	Standard	.900(.001)	.038(.000)	7.1(0.0)	—	.900(.000)	.037(.000)	7.0(0.0)	—
		Classwise	.910(.001)	.066(.000)	17.2(0.1)	—	.903(.000)	.042(.000)	10.3(0.1)	—
		Clustered	.907(.001)	.043(.000)	12.8(0.1)	6	.900(.000)	.029(.000)	7.6(0.0)	100
		CCCP	.900(.001)	**.029(.000)**	7.7(0.1)	101	.900(.000)	**.026(.000)**	7.7(0.0)	101
	APS	Standard	.901(.001)	.081(.001)	58.2(0.8)	—	.900(.000)	.081(.001)	58.2(0.8)	—
		Classwise	.909(.001)	.066(.000)	56.4(0.5)	—	.904(.000)	.042(.000)	52.1(0.6)	—
		Clustered	.902(.001)	.084(.001)	60.0(0.8)	6	.903(.000)	.056(.000)	53.5(0.7)	100
		CCCP	.901(.001)	**.044(.001)**	51.7(0.7)	101	.901(.000)	**.037(.000)**	50.9(0.6)	101
	RAPS	Standard	.901(.001)	.037(.000)	13.1(0.1)	—	.900(.000)	.037(.000)	13.1(0.1)	—
		Classwise	.911(.001)	.066(.000)	23.8(0.1)	—	.903(.000)	.042(.000)	15.5(0.0)	—
		Clustered	.904(.001)	.042(.001)	18.3(0.1)	6	.902(.001)	.032(.000)	13.0(0.1)	100
		CCCP	.901(.001)	**.030(.000)**	12.9(0.1)	101	.901(.000)	**.028(.000)**	12.8(0.1)	101
Food-101 (unbal.)	HPS	Standard	.900(.001)	.063(.001)	6.4(0.1)	—	.900(.000)	.062(.001)	6.3(0.1)	—
		Classwise	.942(.001)	.081(.000)	61.0(0.1)	—	.914(.000)	.053(.000)	19.6(0.2)	—
		Clustered	.896(.001)	.063(.001)	6.5(0.1)	9	.891(.001)	.054(.001)	6.5(0.1)	70
		CCCP	.901(.001)	**.037(.001)**	8.4(0.2)	101	.901(.000)	**.033(.001)**	8.6(0.1)	101
	APS	Standard	.900(.001)	.082(.001)	61.4(0.6)	—	.900(.000)	.082(.001)	61.3(0.7)	—
		Classwise	.943(.001)	.081(.000)	84.2(0.2)	—	.914(.000)	.052(.000)	54.1(0.6)	—
		Clustered	.910(.001)	.080(.001)	62.3(0.6)	9	.910(.001)	.065(.001)	57.9(0.6)	70
		CCCP	.902(.001)	**.049(.001)**	49.5(0.6)	101	.902(.000)	**.041(.001)**	47.2(0.6)	101
	RAPS	Standard	.901(.001)	.040(.001)	12.6(0.1)	—	.900(.000)	.039(.001)	12.5(0.1)	—
		Classwise	.942(.001)	.081(.000)	64.4(0.1)	—	.914(.000)	.053(.000)	25.2(0.2)	—
		Clustered	.903(.001)	.040(.001)	12.7(0.1)	9	.901(.001)	.037(.001)	12.4(0.1)	70
		CCCP	.902(.001)	**.036(.001)**	12.3(0.1)	101	.901(.000)	**.035(.001)**	12.4(0.1)	101
Caltech-256	HPS	Standard	.901(.000)	.077(.001)	2.1(0.0)	—	.900(.000)	.077(.000)	2.1(0.0)	—
		Classwise	.948(.000)	.087(.000)	141.6(0.1)	—	.915(.000)	.058(.000)	7.5(0.1)	—
		Clustered	.891(.001)	.077(.001)	2.2(0.0)	24	.901(.000)	.061(.000)	3.2(0.0)	257
		CCCP	.902(.000)	**.059(.001)**	3.4(0.0)	257	.902(.000)	**.055(.000)**	3.5(0.0)	257
	APS	Standard	.900(.000)	.086(.000)	156.6(1.5)	—	.900(.000)	.085(.000)	156.4(1.4)	—
		Classwise	.948(.000)	.087(.000)	200.9(0.7)	—	.915(.000)	**.058(.000)**	128.3(1.4)	—
		Clustered	.933(.001)	.081(.001)	158.7(1.5)	24	.902(.001)	.068(.000)	126.8(1.5)	257
		CCCP	.901(.000)	**.065(.001)**	123.3(1.5)	257	.901(.000)	.061(.000)	121.5(1.4)	257
	RAPS	Standard	.900(.000)	.069(.000)	13.7(0.1)	—	.900(.000)	.069(.000)	13.7(0.1)	—
		Classwise	.948(.000)	.087(.000)	147.3(0.1)	—	.915(.000)	.058(.000)	17.4(0.1)	—
		Clustered	.923(.001)	.066(.000)	13.8(0.1)	24	.901(.001)	.062(.000)	13.1(0.1)	257
		CCCP	.902(.000)	**.060(.000)**	13.0(0.1)	257	.901(.000)	**.058(.000)**	12.9(0.1)	257
SUN-397	HPS	Standard	.900(.000)	.073(.001)	10.1(0.1)	—	.900(.000)	.073(.001)	10.1(0.1)	—
		Classwise	.945(.000)	.085(.000)	257.3(0.2)	—	.914(.000)	.052(.000)	32.6(0.6)	—
		Clustered	.883(.001)	.072(.001)	9.3(0.1)	89	.897(.001)	.046(.000)	13.9(0.2)	384
		CCCP	.901(.001)	**.044(.000)**	14.6(0.2)	397	.901(.000)	**.043(.000)**	14.7(0.2)	397
	APS	Standard	.899(.001)	.075(.000)	130.3(3.8)	—	.900(.000)	.075(.000)	130.6(3.7)	—
		Classwise	.945(.000)	.085(.000)	308.8(1.3)	—	.914(.000)	.052(.000)	116.1(2.4)	—
		Clustered	.925(.001)	.070(.000)	138.4(4.0)	89	.906(.001)	.051(.000)	105.4(2.6)	384
		CCCP	.900(.000)	**.050(.001)**	97.1(2.5)	397	.901(.000)	**.047(.000)**	94.8(2.2)	397
	RAPS	Standard	.899(.000)	.050(.000)	21.1(0.1)	—	.900(.000)	.050(.000)	21.2(0.1)	—
		Classwise	.946(.000)	.085(.000)	260.4(0.2)	—	.915(.000)	.052(.000)	38.1(0.3)	—
		Clustered	.908(.001)	.048(.000)	21.1(0.1)	89	.902(.001)	.045(.000)	19.9(0.1)	384
		CCCP	.901(.000)	**.044(.000)**	20.1(0.1)	397	.901(.000)	**.043(.001)**	20.5(0.1)	397

Table 3 shows that CCCP achieves the lowest CovGap in almost all scenarios for HPS, APS, and RAPS, especially in the small-calibration regime. At $n_{\text{cal}}/C = 10$, CCCP reduces CovGap by

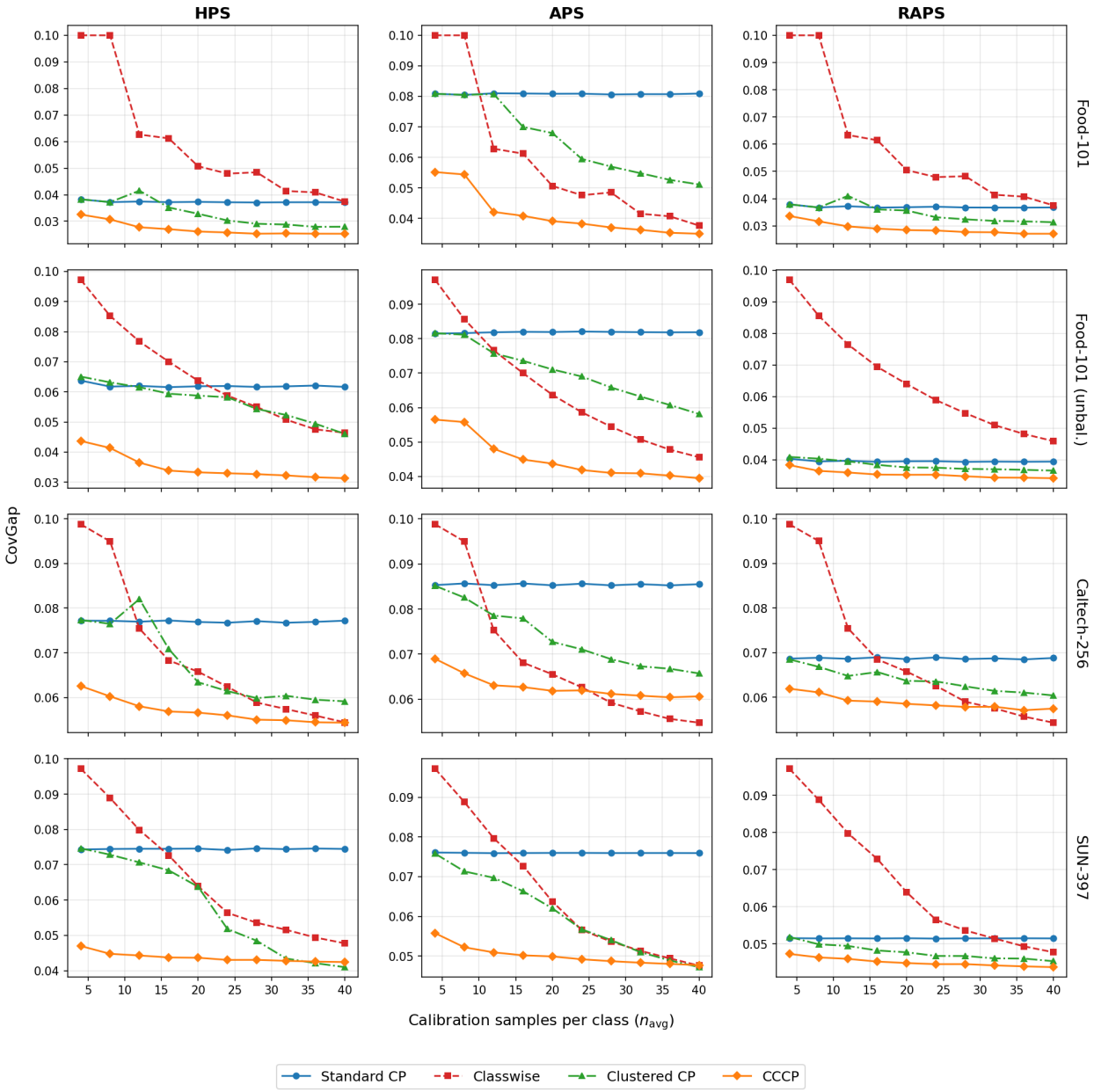


Figure 2: CovGap (mean absolute per-class coverage deviation; lower is better) versus calibration samples per class (n_{cal}/C) for HPS (left), APS (center), and RAPS (right) on each dataset ($n_{\text{cal}}/C =$ 4 to 40 in steps of 2, 20 iterations). Error bars denote ± 1.96 standard errors.

32% on average relative to Standard, 40% relative to Classwise, and 32% relative to Clustered.

Figure 2 shows CovGap across the full sweep of n_{cal}/C. CCCP maintains the lowest CovGap from $n_{\text{cal}}/C = 4$ to 40 on all four datasets. Classwise CP degenerates sharply when $n_{\text{cal}}/C < \lceil 1/\alpha \rceil - 1 = 9$, because classes with fewer than 9 calibration points receive $\hat{q}_y = +\infty$ and trivially enter every prediction set. Clustered CP performs nearly identically to Standard at very low n_{cal}/C: there are so few clustering and calibration points that most classes are classified as rare, which fall back to using the global threshold, effectively reducing the clustered method to the standard conformal prediction method. CCCP avoids both failure modes: because it clusters on

the training set rather than the calibration set, all classes remain non-rare at every n_{cal}/C, and the full calibration budget is available for threshold estimation. As n_{cal}/C grows, Classwise, Clustered, and CCCP all improve. Clustered gradually catches up with CCCP as the γ-split becomes less constraining at larger calibration budgets. When $n_{\text{cal},y}$ becomes sufficiently large for each class to estimate its own threshold reliably, Classwise becomes the best method, as it enjoys the strongest theoretical guarantee of class-conditional coverage. This transition is visible on Caltech-256 at $n_{\text{cal}}/C = 30$, where Classwise achieves the best CovGap under APS.

On the unbalanced Food-101 variant, CCCP's relative advantage is even more pronounced. At $n_{\text{cal}}/C = 30$ with HPS, CCCP reduces CovGap by 47% relative to Standard (.033 vs. .062), compared with 30% on the balanced variant (.026 vs. .037). Class imbalance amplifies the per-class coverage disparity that a single global threshold produces, and CCCP's per-cluster thresholds counteract this effect. CCCP retains all $C = 101$ classes as non-rare under both balanced and unbalanced settings, ensuring that every class receives an adapted threshold regardless of its frequency.

4.3.1 Ensemble CCCP

The cross-fitting procedure in CCCP trains K subfold models. EnsCCCP repurposes these as an ensemble predictor at no additional training cost, yielding higher accuracy and, consequently, smaller prediction sets. Table 4 compares Standard, CCCP, and EnsCCCP at $n_{\text{cal}}/C = 30$ using the HPS score across $K \in \{3, 5, 10\}$ folds.

Table 4: Standard, CCCP, and EnsCCCP with HPS score at $n_{\text{cal}}/C = 30$ ($\alpha = 0.1$, mean over 20 iterations, $n_{\text{train}} = 10{,}000$). Top-1 = test accuracy of the prediction model; Train = total training time (min); Test = test-time inference (s).

Data set	Method	Marg.	CovGap	AvgSize	Top-1 (%)	Train (min)	Test (s)
Food-101	Standard	.900(.000)	.037(.000)	7.0(.0)	60.7(.1)	3.2(0.5)	51(1)
	CCCP (K=3)	.900(.000)	.026(.000)	7.7(.0)	60.7(.1)	8.5(0.7)	51(1)
	CCCP (K=5)	.900(.000)	.026(.000)	7.7(.0)	60.7(.1)	13.9(2.4)	51(1)
	CCCP (K=10)	.900(.000)	.026(.000)	7.7(.0)	60.7(.1)	21.4(1.9)	51(1)
	EnsCCCP (K=3)	.900(.000)	.027(.000)	6.6(.0)	62.2(.0)	5.3(0.5)	157(2)
	EnsCCCP (K=5)	.900(.000)	.026(.000)	5.7(.0)	64.4(.0)	10.7(2.2)	259(1)
	EnsCCCP (K=10)	.900(.000)	.026(.000)	5.1(.0)	65.8(.0)	18.2(1.7)	526(7)
Food-101 (unbal.)	Standard	.900(.000)	.062(.001)	6.3(.1)	62.9(.1)	4.4(0.4)	16(0)
	CCCP (K=3)	.901(.000)	.032(.000)	8.6(.1)	62.9(.1)	11.3(0.6)	16(0)
	CCCP (K=5)	.902(.000)	.033(.000)	8.7(.1)	62.9(.2)	16.2(0.7)	16(0)
	CCCP (K=10)	.901(.000)	.033(.001)	8.6(.1)	62.9(.1)	28.6(0.9)	16(0)
	EnsCCCP (K=3)	.901(.000)	.034(.001)	7.2(.1)	63.5(.1)	7.0(0.4)	47(0)
	EnsCCCP (K=5)	.901(.000)	.034(.001)	6.1(.0)	65.7(.1)	11.8(0.7)	78(1)
	EnsCCCP (K=10)	.901(.000)	.034(.001)	5.5(.0)	67.1(.1)	24.2(1.0)	157(1)
Caltech-256	Standard	.900(.000)	.077(.000)	2.1(.0)	78.3(.0)	3.1(0.1)	6(0)
	CCCP (K=3)	.902(.000)	.056(.000)	3.4(.0)	78.3(.0)	7.7(0.1)	6(0)
	CCCP (K=5)	.902(.000)	.055(.000)	3.5(.0)	78.3(.0)	10.9(0.3)	6(0)
	CCCP (K=10)	.902(.000)	.055(.000)	3.5(.0)	78.3(.0)	17.2(0.2)	6(0)
	EnsCCCP (K=3)	.901(.000)	.057(.000)	3.1(.0)	79.3(.1)	4.6(0.1)	18(0)
	EnsCCCP (K=5)	.902(.000)	.056(.000)	2.7(.0)	80.6(.0)	7.8(0.3)	30(0)
	EnsCCCP (K=10)	.901(.000)	.056(.000)	2.5(.0)	81.5(.0)	14.1(0.2)	59(0)
SUN-397	Standard	.900(.000)	.075(.000)	10.5(.1)	52.7(.0)	4.6(1.0)	47(0)
	CCCP (K=3)	.901(.000)	.043(.000)	15.4(.2)	52.7(.0)	9.3(1.0)	47(0)
	CCCP (K=5)	.901(.000)	.043(.000)	15.4(.1)	52.7(.0)	12.7(0.9)	47(0)
	CCCP (K=10)	.901(.000)	.043(.000)	15.4(.2)	52.7(.0)	20.4(1.0)	47(0)
	EnsCCCP (K=3)	.901(.000)	.046(.000)	11.5(.0)	53.1(.1)	4.7(0.2)	144(1)
	EnsCCCP (K=5)	.901(.000)	.045(.000)	9.5(.0)	55.8(.0)	8.1(0.2)	239(2)
	EnsCCCP (K=10)	.901(.000)	.045(.000)	8.3(.0)	57.5(.0)	15.8(0.2)	483(3)

EnsCCCP achieves higher accuracy than the single-model methods, a direct consequence of the

ensemble's improved predictions. This accuracy gain translates to substantially smaller prediction sets: on Food-101 at K=10, EnsCCCP reduces AvgSize from 7.7 (CCCP) to 5.1 while maintaining the same CovGap (.026). On Caltech-256, Standard has the smallest set size because HPS assigns most probability mass to one or two classes, so the global threshold already produces small sets; however, Standard's CovGap (.077) is far worse than EnsCCCP's (.056). EnsCCCP pairs the ensemble's accuracy gain with CCCP's class-aware thresholding, keeping CovGap within .003 of CCCP on every dataset. On the unbalanced variant at K=10, EnsCCCP achieves a CovGap of .034 while reducing AvgSize from 8.6 (CCCP) to 5.5.

CCCP's CovGap and AvgSize are virtually invariant to K: across all datasets, CovGap varies by at most .001 and AvgSize by at most 0.1, confirming that the cross-fitting procedure depends on the training-set structure rather than the fold count. In contrast, EnsCCCP benefits from larger K because more subfold models produce a stronger ensemble. From $K = 3$ to $K = 10$, top-1 accuracy improves by 2.2 percentage points on Caltech-256, 3.6 on Food-101, 3.6 on Food-101 (unbal.), and 4.4 on SUN-397. This accuracy gain translates to proportionally smaller prediction sets; for example, AvgSize on Caltech-256 drops from 3.1 to 2.5 (-19%). Crucially, EnsCCCP's CovGap remains stable across K, varying by at most .001, so the accuracy and set-size improvements come at no cost to conditional coverage. Since training cost scales linearly with K, we recommend $K = 10$ when computational budget permits.

5 Conclusion

Cross-fitted clustered conformal prediction removes the data trade-off inherent to clustered conformal prediction by extracting clustering information from the training set via cross-fitting, so that the full calibration budget serves calibration while the typically much larger training set supplies the clustering step. Because the clustering map depends only on the training data, CCCP provides clean cluster-conditional coverage without validity subtleties. On three real-world image-classification datasets (plus a controlled unbalanced variant), CCCP attains the lowest class-conditional coverage gap among all methods tested. The ensemble variant EnsCCCP further reduces prediction-set size by repurposing the subfold models as a stronger predictor at no additional training cost.

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A Notation

Table 5: Summary of notation.

Symbol	Meaning
C	number of classes
f	prediction model (softmax classifier)
$f_y(x)$	softmax output of f for class y at input x
$s(x, y)$	nonconformity score at class y given input x
α	target miscoverage level
$\mathcal{C}_\alpha(x)$	prediction set for input x at level α
$\hat{q},\ \hat{q}_y,\ \hat{q}_m$	marginal, classwise, and clusterwise calibration thresholds
$\mathcal{D}_{\text{train}},\ \mathcal{D}_{\text{cal}}$	training and calibration sets
n_{cal}	calibration set size ($\lvert\mathcal{D}_{\text{cal}}\rvert$)
$n_{\text{cal},y}$	number of calibration points in class y
M	number of clusters
$\hat{h}(y)$	cluster assignment function (class $\to$ cluster)
K	number of cross-fitting folds
$\mathcal{F}_k$	k-th cross-fitting fold
g_k	subfold model trained on $\mathcal{D}_{\text{train}} \setminus \mathcal{F}_k$
f_{ens}	ensemble predictor ($\frac{1}{K}\sum_k g_k$)